EEG-based Motor Imagery Signal Classification for BCI-Controlled Assistive Robot Receptionist Systems

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INTRODUCTION

Individuals with severe mobility impairments face challenges using conventional computer interfaces, limiting their participation in service roles. Brain-Computer Interface (BCI) technology offers an alternative through direct neural control. This study utilized an EEG-based motor imagery (MI) BCI framework for left-right control of an assistive robot receptionist. To ensure accessibility, we employed the consumer-grade Emotiv EPOC X headset. Motor execution (ME) was integrated into MI training to enhance neural discriminability, particularly for users with limited MI proficiency. EEGNet and Support Vector Machine (SVM) algorithm were compared, and the most effective model was selected to train a personalized AI model for six participants, which were then integrated with User Interface (UI). At last, the system was evaluated. **If reliable control is achieved in healthy individuals, the framework may be extended to support real-world assistive applications.**

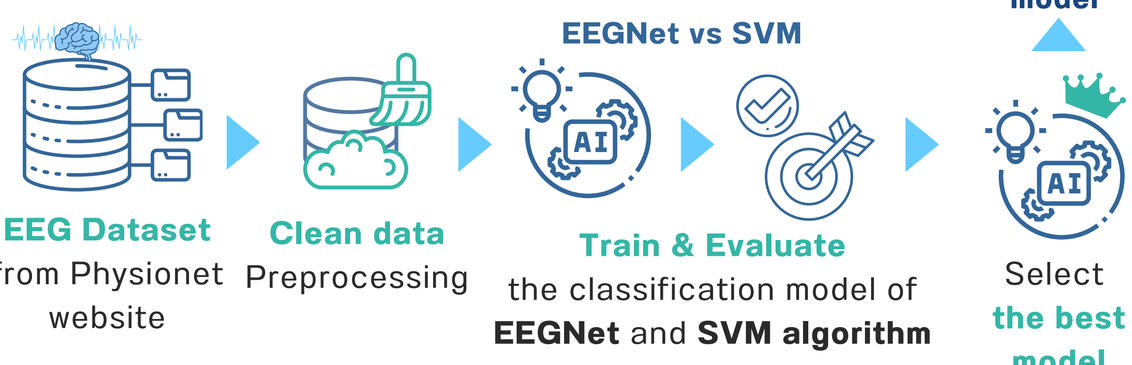


OBJECTIVE

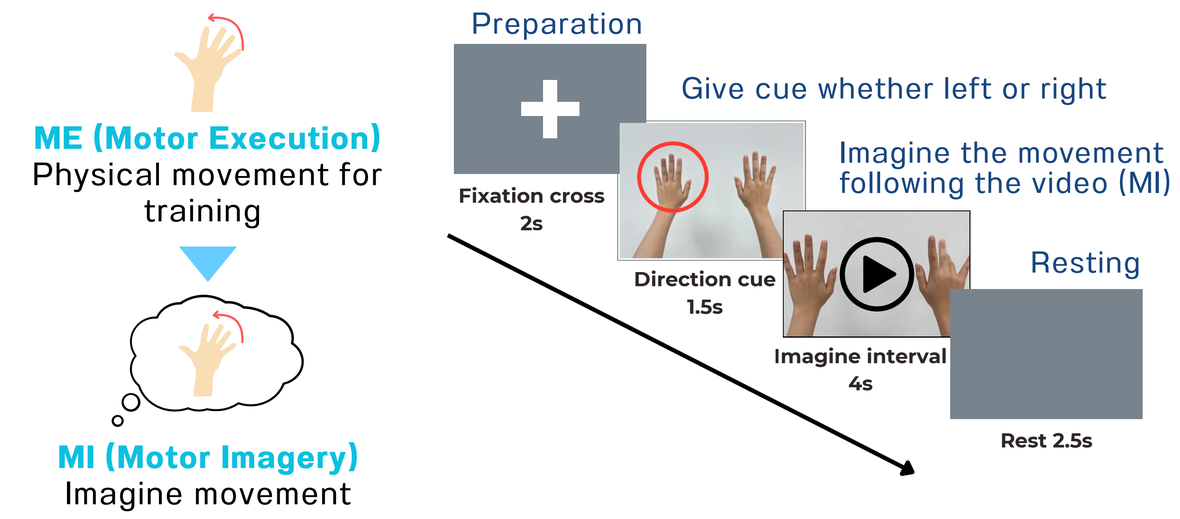
- To **develop and determine the most effective model for classifying EEG signals** associated with left and right motor imagery.
- To **evaluate the effectiveness of the BCI-controlled** assistive robotic receptionist interface in terms of **usability and user feedback**.
- To **design and implement a BCI-controlled** assistive robotic receptionist **interface**.

METHODOLOGY

1. Model Preparation

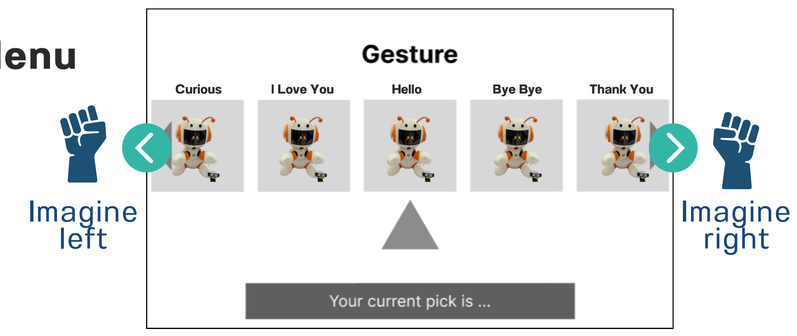


2. Experimental Protocol



3. User Interface Menu

To pick gestures of the receptionist robot using EEG MI signal.



4. Attitudinal and Behavioral Evaluation (User Feedback and Usability)

User Feedback



Usability

- Task success rate on UI Testing

RESULTS AND DISCUSSION

1. Model Preparation

Table 1 The comparison of EEGNet and SVM algorithm

	Accuracy	Confusion Matrix (Wrong : Correct)	Log Loss	Computational Efficiency
EEGNet — Deep Learning	71.99 %	1 : 2.6	0.60	Very Complex • Delay : 150 ms
SVM — Supervised Machine Learning	62.69 %	1 : 1.7	0.69	Less Complex • Delay : 1 ms

EEGNet was selected as the classifier in the personalized model implementation of the robot receptionist interface for its **higher accuracy and acceptable latency** (Table 1).

2. Personalized Model & UI Performance

Personalized EEGNet performance across six participants was classified into two groups: **high accuracy (>0.70)** and low accuracy (~0.55) (Fig 1, blue bar). The Real-Time decoding correctness averaged 0.55, reflecting limited reliability (Fig 1, red bar).

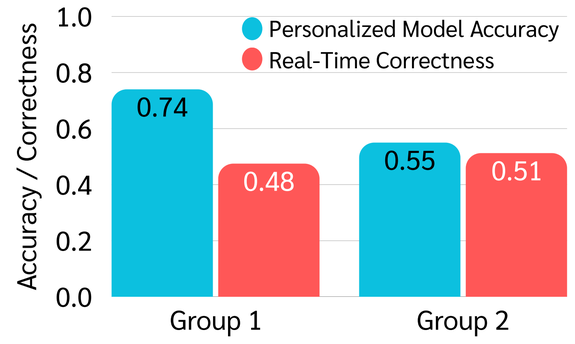


Figure 1 Personalized Model Accuracy and Real-Time Correctness

Table 2 UI Testing: Task Success Rate

Group	Success Rate
Group 1	0.2
Group 2	Impractical model

Correspondingly, the UI task success rate was 0.2 for one participant (Table 2). The users' attitudes toward the UI experience revealed **high comfort with usability feedback** (4.13), moderate satisfaction with system performance (3.75), and low perceived command accuracy (3.08).

3. Emotiv EPOC X & the User's MI Ability Limitation

Performance was affected by **the EEG signal quality** and **the user's motor imagery (MI) ability**. The Emotiv EPOC X's **fixed electrodes** and variable fit reduced signal quality, while some users struggled with MI tasks.

CONCLUSION

- **The EEGNet model was selected for its higher accuracy (71.99 %) with acceptable latency (150 ms)**, though real-time performance remained limited.
- Results varies **across users: personalised model accuracy (>70%, 50%)** influenced by MI ability and EEG signal quality.
- While the system is feasible, improvements in signal acquisition, user training, and data size are needed to achieve reliable real-time control and BCI evaluation.



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